CODEALPHA INTERNSHIP

EDUCATIONAL PERFORMANCE & RESOURCE ALLOCATION

Power BI Dashboard Project Report

Task 4 • Data Analytics / Business Intelligence

Prepared by
Manya
B.Tech – Information Technology

Professional documentation covering project objectives, methodology, dashboard design, KPIs, visual analysis, insights, recommendations, limitations, and future enhancements.

1. Executive Summary

This project presents an interactive, single-page Power BI dashboard developed as part of the CodeAlpha Internship, Task 4: Educational Performance and Resource Allocation. The solution brings together academic performance, attendance, dropout indicators, teacher resources, budget utilization, grants, and digital infrastructure usage into one analytical view.

The dashboard is designed to move from high-level monitoring to focused district- and institution-level analysis. KPI cards provide a quick summary, while comparative charts, a time trend, ranking analysis, and slicers support deeper exploration.

The project demonstrates practical skills in data preparation, data modeling, DAX measure creation, visualization, dashboard design, and communication of data-driven insights.

Project objective

To transform educational data into a clear, decision-oriented analytical dashboard that can help identify performance patterns, resource distribution differences, and areas requiring further investigation.

Project	Educational Performance & Resource Allocation Dashboard
Organization	CodeAlpha
Task	Task 4 – Educational Performance and Resource Allocation
Domain	Data Analytics / Business Intelligence
Primary Tool	Microsoft Power BI
Author	Manya – B.Tech, Information Technology
Format	Single-page interactive dashboard

2. Task Requirements & Solution Mapping

The dashboard was structured to address the core CodeAlpha task requirements through performance monitoring, resource analysis, and interactive exploration.

Requirement	Dashboard solution
Visualize academic performance metrics	Average GPA, attendance, dropout rate, GPA by district, and GPA trend provide complementary views of academic performance.
Analyze resource usage and identify potential gaps	Teacher distribution, budget utilization, grants, and top institutional internet usage provide resource and infrastructure perspectives.
Support data-driven education decisions	KPI monitoring, district comparisons, institution filtering, and trend analysis help users identify areas for further review.

Design principle: With a one-page constraint, the final layout prioritizes the most decision-relevant KPIs, five core charts, and a limited set of slicers rather than overcrowding the canvas.

3. Methodology & Analytical Workflow

Data understanding — Reviewed datasets and identified fields relevant to students, institutions, teachers, infrastructure, and resources.

Data preparation — Prepared relevant fields for analysis and ensured numeric, categorical, and time dimensions were usable in Power BI.

Data modeling — Worked with available tables and relationships so relevant indicators could be analyzed together.

DAX analysis — Created measures for KPIs and analytical indicators such as GPA, attendance, dropout rate, teachers, grants, and resource utilization.

Visualization — Selected KPI cards, comparative charts, trend analysis, and ranking visuals according to the analytical question.

Interactivity — Added slicers for district, year, and institution where the underlying model supports the filter context.

Dashboard design — Built a professional single-page layout with a consistent purple theme and clear visual hierarchy.

Insight generation — Reviewed KPI values and visual patterns to identify observations and areas for deeper analysis.

4. KPI Framework

KPI	Dashboard value	Purpose
Total Students	210M	Scale of the student population represented in the dashboard.
Average GPA	3.00	High-level academic performance indicator.
Average Attendance	80.02%	Student participation indicator.
Average Dropout Rate	3.55%	Student retention indicator.
Total Teachers	167K	Overall faculty resource indicator.
Average Grant	4.94M	High-level financial/resource support indicator.

Values reflect the supplied final dashboard view; K and M follow the dashboard display convention.

5. Dashboard Visualizations

1. Average GPA by District

Compares academic performance across districts and highlights relative differences.

2. Top 5 Institutions by Average Internet Usage

Ranks the five institutions with the highest average internet usage, keeping the visual focused and readable.

3. Average Budget Utilization by District

Compares district-level budget utilization and supports review of resource use.

4. Average GPA Trend

Tracks GPA over time and helps identify stability or changes in academic performance.

5. Total Teachers by District

Compares faculty distribution across districts and supports resource analysis.

6. Final Dashboard Preview

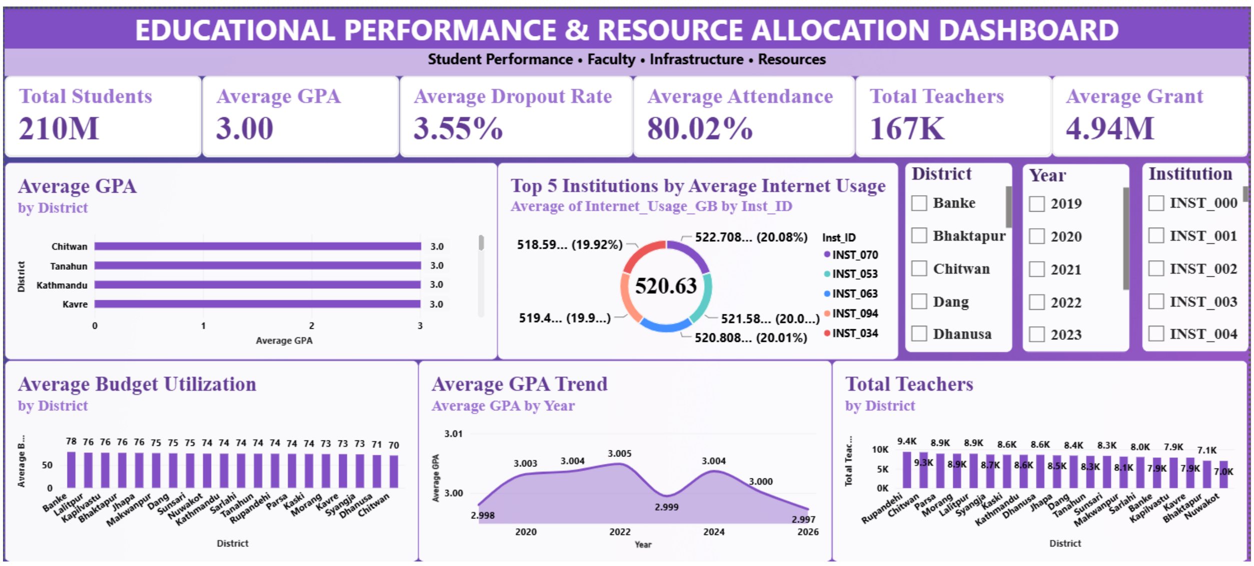

Figure 1. Final single-page Power BI dashboard supplied for this report.

7. Interactivity & User Experience

District — Supports focused comparison of district-level academic and resource indicators.

Year — Supports time-based filtering and complements the GPA trend analysis.

Institution — Supports institution-level exploration, particularly for infrastructure-related analysis.

8. Key Insights

• Average GPA is approximately 3.00, providing a baseline for monitoring academic performance.

• Average attendance is approximately 80.02%, making participation an important indicator alongside academic outcomes.

• Average dropout rate is approximately 3.55%, providing a useful student-retention signal for further investigation.

• Teacher availability varies across districts, indicating differences in faculty distribution.

• Budget utilization varies across districts and can be reviewed against local educational needs.

• The top-five internet-usage view provides a focused comparison of institutions without overwhelming the dashboard.

• Combining academic, faculty, financial, and infrastructure indicators provides broader decision context than a single metric alone.

9. Recommendations

• Investigate districts with comparatively lower academic performance or weaker resource indicators.

• Compare teacher distribution with student population and student-teacher ratio before staffing decisions.

• Review districts with lower budget utilization to distinguish planning, execution, reporting, or genuine underspending.

• Monitor attendance and dropout together to identify areas requiring deeper student-retention analysis.

• Use institution-level infrastructure indicators to prioritize digital-resource improvements where appropriate.

• Repeat the analysis periodically so changes over time can be monitored.

10. Technical Implementation

The dashboard was developed in Microsoft Power BI using data preparation, modeling, DAX measures, and interactive visual design.

Core measures include:

• Total Students

• Average GPA

• Average Attendance

• Average Dropout Rate

• Total Teachers

• Average Student-Teacher Ratio

• Average Budget Utilization

• Average Grant

• Infrastructure-related indicators such as internet usage and electricity stability where available

11. Data Quality & Interpretation Notes

Dashboard values should be interpreted in the context of the source dataset and its available dimensions. Where a metric shows very small category differences, bars or points may appear visually similar even when the underlying values differ slightly. A measure can only respond to a slicer when the relevant tables and relationships support that filter context.

For production deployment, additional validation should include missing-value checks, duplicate checks, range validation, relationship validation, and reconciliation of aggregates against the source system.

12. Limitations

This dashboard is an internship/project prototype. It does not establish causal relationships or prescribe resource allocation without additional operational context. Financial, staffing, infrastructure, and performance indicators should be validated against current institutional records before real-world decisions are made.

13. Future Enhancements

• Add drill-through pages for institution and district investigation.

• Introduce richer time-series analysis for attendance, dropout, and resource indicators.

• Add targets or benchmarks to compare actual performance against expected performance.

• Develop a resource-gap score combining staffing, budget, and infrastructure indicators.

• Add automated refresh and governed production data sources.

• Create richer tooltip pages for additional context.

14. Project Deliverables

Deliverable	Purpose

README.md	Project overview, methodology, visuals, insights, and documentation.
Power BI .pbix	Editable source dashboard containing the model, measures, visuals, and report layout.
dashboard.png	Final dashboard preview for GitHub and portfolio presentation.
Dataset	Source data used for the analysis, subject to sharing/licensing constraints.
Project Report PDF	Professional documentation of the project and analytical approach.

15. Internship Context

This project was developed as part of the CodeAlpha Internship in the Data Analytics / Business Intelligence domain. It provided practical experience in converting educational data into an interactive business-style dashboard, with emphasis on analytical thinking, visualization, and communication of insights.

16. Conclusion

The Educational Performance & Resource Allocation Dashboard demonstrates how Power BI can transform multiple educational indicators into a centralized, interactive analytical solution. By combining student outcomes, attendance, dropout, teacher resources, budget utilization, grants, and infrastructure usage, the project provides a practical foundation for monitoring performance and identifying areas for deeper analysis.

Prepared by: Manya • B.Tech – Information Technology • CodeAlpha Internship – Task 4 • Data Analytics / Business Intelligence